



RIANNON K. COLTON

rcolton@uark.edu

EDUCATION

Doctor of Philosophy in Geosciences: University of Arkansas. Fayetteville, AR. May 2029.

Advisor: Dr. Matthew Covington.

Bachelor of Science in Earth & Atmospheric Sciences: Georgia Institute of Technology. Atlanta, GA. Dec 2019.

EXPERIENCE

University of Arkansas Department of Geosciences:

- **Graduate Teaching Assistant:** Jan 2023 – Present.
 - **Quantitative Methods in Earth Science:** Aug 2024 – Present. Grade lab assignments, hold office hours, and mentor undergraduate and graduate students in data analysis methods using Python.
 - **Earth Science Lab:** Aug 2025 – Present. Teach and mentor undergraduate students about different aspects of Earth Science, grade lab assignments, and hold office hours.
 - **Physical Geology Lab:** Jan 2023 – Present. Run multiple lab sections of Physical Geology Lab, teach and mentor undergraduate students in different aspects of geology, create and grade assignments, and hold office hours.
- **Graduate Research Assistant:** Apr 2025 – May 2025.
 - **Understanding the variability in radon and carbon dioxide in Carlsbad Cavern:** Design, construct, deploy, maintain, and conduct fieldwork with speleometeorological stations in Carlsbad Caverns National Park.
- **Carbonate Critical Zone Research Coordination Network:** May 2023 – May 2024.
 - **Rates and Times Working Group:** Conduct a literature review with a group of researchers to develop a review paper on topics related to rates and time in carbonate systems.
 - **Arkansas Field Workshop:** May 2023. Organize and run student training in data collection and analysis, mentor undergraduate students from different universities, and advance the development of review papers and other outputs.

Buffalo National River:

Firefighter Type 2 (FFT2): Jan 2024 – Present.

- Serve on a hand crew and engine crew, performing fire suppression and fuels management duties in the most adverse climate, fuel, and terrain conditions.

Geoscience/Biology Assistant: *Jul 2020 – Dec 2020.*

- **Analyze how White Nose Syndrome affects bat usage relative to vegetation type, season, temperature, and proposed management activities:** Design bat detector deployments, collect metadata on each deployment, and process echolocation files through Kaleidoscope Pro.
- **Examine gain/loss behavior in the Big Creek Watershed:** Assist the USGS with placing charcoal packets to determine water flow and a dye trace injection.
- **Ground truth park sections to detect undocumented caves/karst features:** Assist the Cave Research Foundation and other organizations with documenting cave/karst resources, and analyze data using LiDAR and GIS.
- **Map soil thickness over the bedrock surface to aid wastewater treatment system installations:** Assist the USGS with Electric Resistivity Tomography (EMT) surveys.

Oregon Caves National Monument: Natural Resources Management Assistant. *May 2025 – Aug 2025.*

- **Cave climate monitoring:** Monitor CO₂, temperature, humidity, airflow speed and direction, air quality, and other environmental factors using Wöhler devices, HOBO devices, and PurpleAir air quality monitors to determine current conditions in Oregon Caves.
- **Cave management:** Help mitigate human impacts along established tour routes, manage caves/caving gear, and oversee volunteers/volunteer projects.

Willamette National Forest, Middle Fork Ranger District, United States Forest Service (USFS): Hydrologic Technician. *May 2024 – Aug 2025.*

- **Survey logging units to evaluate road accessibility and hydrologic features:** Ground truth logging units to detect streams using LiDAR tracks, map and classify these streams, and deploy temperature loggers in streams. Survey roads for length, culvert conditions, and blockages including berms, down trees, and water bars within these logging units, and analyze accessibility and engineering costs.
- **Assist other Natural Resources departments with various projects:** Assist the Fisheries Department with placing fish antennae for tracking bull trout and deploying and maintaining caddisfly traps, assist the Soils Department with soil compaction detection using a penetrometer, assist the Archaeology Department with monitoring archaeologically significant rock shelters by searching for artifacts and new possible impacts and changes to the sites, and assist the Wildlife Department with arboreal climbing training.
- **Wildfire Natural Resource Advisor (READ):** Work with Incident Management Teams and Fireline personnel to minimize the impacts of wildland fire and fire management actions on natural, cultural, and social resources.

Provide critical support for repairing fire suppression damages and identifying and mitigating threats caused by wildfires.

Carlsbad Caverns National Park: Caves and Karst Assistant. *Jan 2022 – Sep 2022.*

- **Carlsbad Cavern Carbon Dioxide Preliminary Analysis 2019-2022:** Calculate visitor contribution to carbon dioxide levels in Carlsbad Caverns by collecting data with Wöhler devices and modifying R Script analysis scripts.
- **Cave climate monitoring:** Monitor CO₂, temperature, humidity, radon, and other environmental factors using Wöhler devices, psychrometers, radon monitors, and Traftx to determine current conditions in Carlsbad Caverns.
- **White Nose Syndrome detection:** Collect acoustic bat monitor data and process files through SonoBat, and assist with mist-netting and guano/sediment sampling for the detection of Pd.
- **Ozone monitoring:** Operate a 30-foot-tall weather tower to collect wind speed, relative humidity, temperature, solar, and rainfall data.
- **Cave management:** Help mitigate human impacts along established tour routes, manage caves/caving gear, and oversee volunteers/volunteer projects.
- **Carlsbad Caverns Search and Rescue Team:** Respond to medical emergencies and missing persons.

Jewel Cave National Monument: Caves and Karst Assistant. *May 2021 – Oct 2021.*

- **Correlate surface and subsurface geology:** Surface field mapping, caving trips to inventory and map subsurface geologic features and document field observations.
- **Cave management:** Help mitigate human impacts along established tour routes, complete routine microclimate monitoring, collect acoustic bat monitor data and process files through SonoBat, and assist with data collection for an ongoing hydrology project.

Georgia Institute of Technology:

Undergraduate Research:

- **Spatial Ecology and Paleontology Lab: *Aug 2017 – May 2018.*** Research how intraspecific variation in *Microtus* changed in the Southwest USA from 20,000 - 3,000 years ago.
 - *Responsibilities:* Landmark specimen photographs and compare landmark locations.
- **Space Systems Design Lab: *Jan 2016 – Dec 2016.*** Student-run mission (TARGiT) funded by NASA USIP to remotely sense surface topography of planetary bodies in high precision from a CubeSat platform using tethering and ranging (LiDAR).
 - *Responsibilities:* Construct science goals, science requirements, and satellite applications.

- On Dec 21, 2021, TARGiT was launched aboard the SpaceX Cargo Dragon and deployed from the International Space Station on Jan 26, 2022.
- **South China Sea Research Cruise:** *Jun 2016*. Measure nutrient fluxes to the South China Sea with an international ocean science team on the Schmidt Ocean Institute research vessel R/V Falkor.
 - *Responsibilities:* Collect water from a CTD, conduct XBT drops, process PNs soda cans, tow plankton nets, and blog regarding daily ship life.
- **Ferrier Surface Processes Group:** *May 2015 – May 2016*. Tectonic influences on chemical erosion rates.
 - *Responsibilities:* Use x-ray spectrometer (XRF) to analyze chemical makeup of crushed rock and soil samples.

LEADERSHIP AND TEAMWORK

Volunteering:

- **United States Deep Caving Team (USDCT):** *April 2025 – Present*. Expedition survey and exploration in Sistema Cheve, Oaxaca, Mexico.
- **Southwestern Region (SWR) Cave Rescue:** *Mar 2024 – Present*. On call for cave rescues in the Southwest United States.
 - **Parks Ranch Cave Rescue:** March, 2025.
- **National Cave and Karst Research Institute (NCKRI):** *Mar 2022 – May 2023*. Assist with conducting Electric Resistivity Tomography surveys, create science content for the Caves and Karst word of the week project, and inventory, database, and curate gear and geology samples.
- **National Park Service (NPS):** *Jul 2021 – Present*. Discover, explore, inventory, and map caves and karst features and participate in cave restoration projects and expeditions at Jewel Cave National Monument, Wind Cave National Park, Carlsbad Caverns National Park, Oregon Caves National Monument, and Grand Canyon National Park.
- **United States Forest Service (USFS):** *Jul 2021 – Present*. Discover, explore, inventory, and map caves and karst features and participate in cave restoration projects and expeditions at Lincoln National Forest, Siskiyou National Forest, and Bighorns National Forest.
- **National Speleological Society (NSS):** *Jan 2022 – Jun 2022*. 2022 NSS Convention Volunteer Coordinator.
- **Iceland Search and Rescue (ICE-SAR):** *Jun 2018 – Aug 2018*. Train search and rescue dogs.

Cave Research Foundation (CRF):

- **Joint Venturer:** *Aug 2020 – Present.* Discover, inventory, and map caves for natural areas including Arkansas Natural Heritage Commission land, The Buffalo National River, and Mark Twain National Forest.
- **Fellow:** *Nov 2025 – Present.*

Professional Associations:

- **National Speleological Society (NSS):** *June 2017 – Present.*
- **Geological Society of America (GSA):** *January 2023 – Present.*
- **Association for Women Geoscientists:** *Sep 2023 – Sep 2024.* Ozark Hawgs Chapter.

Speleological Society Associations:

- **Boston Mountain Grotto:** *Jan 2023 – Present.* Member.
- **Pecos Valley Grotto:** *Mar 2024 – Present, Jan 2022 – Dec 2022.* Member.
- **Oregon Grotto:** *May 2024 – May 2025.* Member.
- **Pahasapa Grotto:** *May 2021 – May 2022.* Member.
- **Alabama Cave Survey:** *2020.* Member.
- **Southeastern Cave Conservancy (SCCI):** *2020.* Member.
- **Hawai'i Speleological Survey:** *2019.* Member.
- **Hawai'i Grotto:** *May 2019 – May 2022.* Member.

GRANTS

Western National Parks Association 1-Year Research Grant: *2024 - 2025.*
Award Amount: \$19,782. PI: Dr. Matthew Covington. Co-PI: Riannon Colton.

HONORS AND AWARDS

PhD Fellowship in Geosciences: University of Arkansas Fullbright College of Arts & Sciences. *Aug 2025.*

Jack D. Mussett Scholarship: University of Arkansas Fulbright College of Arts & Sciences. *Fall 2024.*

Mark and Theresa Gentry Land and Water Resources Scholarship: University of Arkansas Department of Geosciences. *May 2024.*

W. Joe Edwards, JR. Geology Scholarship: University of Arkansas Department of Geosciences. *May 2024.*

Miss Congeniality: Nicest GEOS Student (2023-2024): University of Arkansas Department of Geosciences. *May 2024.*

30th Anniversary of the International Karstological School: Research Centre of the Slovenian Academy of Sciences and Arts (ZRC SAZU) and UNESCO. *June 2023*. To diffuse karstological knowledge and promote research and exploration.

Bridges Undergraduate Research Award: Department of EAS at Georgia Tech. *Mar 2016*. To undertake a research project under the mentorship of a Georgia Tech EAS faculty.

Instructor of the Year: Outdoor Recreation at Georgia Tech. *Apr 2018*.

TRAININGS AND CERTIFICATIONS

SDI Open Water SCUBA Diver: SDI TDI ERDI Certified.

First Aid: *June 2024*. Oregon Emergency Medical Trainers.

CPR/A.E.D.: *June 2024*. Oregon Emergency Medical Trainers.

National Cave Rescue Commission (NCRC) Small Party Assisted Rescue (SPAR): *March 2024*.

Intro to Cave Rescue: *June 2021*. Wind Cave National Park.

PUBLICATIONS

Colton, R., and Covington, M.D., 2025, Understanding the variability in gas concentrations of Carlsbad Cavern, 19th International Congress of Speleology and 38 Congresso Brasileiro de Espeleologia, Vol. V.

Miller, B.V., Bitting, C.J., and **Colton, R.**, 2022, Seepage investigation and dye tracing to characterize base flow stream behavior in Big Creek watershed, Newton County, Arkansas: U.S. Geological Survey data release, <https://doi.org/10.5066/P97T4C52>.

Adams, R.F., Miller, B.V., Bitting, C.J., Boyles, Z., **Colton, R.**, and Kress, W. H., 2022, Electric Resistivity Tomography (ERT) surveys conducted near campgrounds on the Buffalo National River, Arkansas: U.S. Geological Survey data release, <https://doi.org/10.5066/P9CRLINN>.

PRESENTATIONS

Colton, R., and Covington, M.D. (2025). Understanding the variability in gas concentrations of Carlsbad Cavern [Oral]. Presented at the 2025 Geological Society of America (GSA) Connects Meeting, San Antonio, Texas, USA.

Colton, R., and Covington, M.D. (2025). Understanding the variability in gas concentrations of Carlsbad Cavern [Oral]. Presented at the 19th International Congress of Speleology and 38 Congresso Brasileiro de Espeleologia, Belo Horizonte, Minas Gerais, Brazil.

Colton, R., and Covington, M.D. (2025). Understanding the variability in radon and carbon dioxide in Carlsbad Cavern [Oral]. Presented at the 2025 Geological Society of America (GSA) South-Central Geographic Section Meeting, Conway, Arkansas, USA.

Colton, R., and Covington, M.D.. (2024). Understanding the variability in radon and carbon dioxide in Carlsbad Cavern [Poster]. Presented at the University of Arkansas, Department of Geosciences GeoHog Conference 2024, Fayetteville, Arkansas.

Colton, R., and Covington, M.D.. (2023). Cave airflow mechanism and ventilation controls over radon-222 dynamics [Poster]. Presented at the 30th Anniversary of the International Karstological School, Postojna, Slovenia.

Cromwell, L., Baraza, T., Jones, M., Oleson, E., **Colton, R.**, Carton De Grammont, P., Covington, M., and Martin, J.. (2023). From field to file: promoting diversity within critical zone science through undergraduate student trainings [Oral]. Presented at GSA Connects 2023, Pittsburg, Pennsylvania.

Miller, B. V., **Colton, R.**, Bitting, C.J.. (2023). Seepage investigation and dye tracing in Big Creek watershed near Buffalo National River, Arkansas [Oral]. Presented at the 25th National Cave & Karst Management Symposium, Chattanooga, Tennessee.

Colton, R., and Covington, M.D.. (2023). Cave airflow mechanism and ventilation controls over radon-222 dynamics [Poster]. Presented at the University of Arkansas, Department of Geosciences GeoHog Conference 2023, Fayetteville, Arkansas.